



Republic of the Philippines
DAVAO ORIENTAL STATE UNIVERSITY
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Information Technology Program

**VOTECHAIN: A BLOCKCHAIN-INTEGRATED WEB-BASED VOTING SYSTEM
FOR SECURE AND TRANSPARENT STUDENT ELECTIONS AT DAVAO
ORIENTAL STATE UNIVERSITY**

ITC 130
Application Development in Emerging Technologies
Project Proposal

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FEBRUARY 2026

I. Proposed Title

VoteChain: A Blockchain-Integrated Web-Based Voting System for Secure and Transparent Student Elections at Davao Oriental State University

II. Proponents / Researchers

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III. Program / Year / Section

BS Information Technology – 3rd Year – A

IV. Project Type

- (✓) Research
- (✓) Capstone Project
- (✓) System Development
- () Mobile Application
- (✓) Web Application

V. Introduction / Background of the Study

School Elections are an essential part of academic institutions, as they promote leadership, democracy, and student representation. Traditional voting methods, whether paper-based or simple electronic voting systems, may encounter issues in terms of security, transparency, and accuracy of the results. Manual voting processes are vulnerable to issues such as ballot loss, human error, and delays in vote counting. Similarly, traditional electronic voting systems that do not utilize advanced security mechanisms can be vulnerable to ballot tampering, vote manipulation and insider threats. These issues may cause voters to doubt the results (Aswini, 2025). Because of this there is a need for a more secure, reliable, and trustworthy voting system, even within academic institutions.

The advancement of blockchain technology provides an opportunity to improve the security, transparency, and reliability of digital voting systems.

Blockchain technology uses a decentralized and secure ledger to record votes, ensuring they cannot be tampered with or deleted. Studies show that blockchain-based voting systems can improve data integrity, transparency, and resistance to unauthorized changes compared to traditional electronic voting systems (Singh et al., 2022; Dhruthana et al., 2025). However, most existing studies focus on large-scale elections and do not fully address the specific needs of school-level elections, such as ease of use, web accessibility, and the integration within a university environment. To address this gap, this study proposes VoteChain, a blockchain-integrated web-based voting system designed to provide secure authentication, transparent vote recording, and accurate automated vote counting. The system aims to enhance the integrity, efficiency and trustworthiness of student elections at Davao Oriental State University by providing a modern, secure and user-friendly digital voting platform.

VI. Objectives of the Study

General Objective

To develop a blockchain-integrated web-based voting system that ensures secure, transparent, and reliable student elections at Davao Oriental State University.

Specific Objectives

1. To develop a secure and user-friendly web-based voting platform
2. To integrate blockchain technology for secure vote recording and automated counting
3. To evaluate the system's usability, performance, and reliability for school-level elections

VII. Target Users / Beneficiaries

- DOrSU - Commission on Student Election
- Student Organizations
- Student Voters

VIII. Scope and Limitations

Scope

- The system will be a web-based voting platform accessible to students and election administrators.
- It will integrate blockchain technology for secure and tamper-resistant vote recording.
- The system will allow student authentication and eligibility verification
- Votes will be automatically counted, and election results will be generated accurately
- Designed for school-level elections at DOrSU-Maiin, including student organizations (Student council, Faculty-based and program-based orgs)

Limitations

- It will not include mobile applications, voting is only via web browser
- The system depends on internet connectivity for access and voting, offline voting is not supported
- The system will be developed exclusively for student elections at Davao Oriental State University.
- The blockchain feature will be implemented for secure vote recording within the system and will not involve deployment on a public blockchain network.

IX. Proposed Technology / Tools to be Used

Software:

- Frontend: HTML, CSS, Javascript
- Backend: PHP
- Database: Supabase (PostgreSQL-based)
- Blockchain Integration: Custom Lightweight Blockchain (Hash)

Hardware:

- Laptop

X. Conceptual Framework / System Overview

The system begins when student voters log in using their authorized university credentials. Once verified, the system identifies the election level they are eligible to vote in (University, Organization, or Program). The voter selects their preferred candidates and submits their vote through the web interface.

After submission, the system validates the vote to ensure the student has not voted more than once. The vote data is then encrypted and recorded in the blockchain ledger. Because blockchain records are tamper-resistant and time-stamped, the stored votes cannot be altered without detection.

The system automatically tallies all recorded votes and generates election results that can be viewed by authorized election officers. Through this process, VoteChain ensures data integrity, transparency, and reliability in managing student elections.

Input:

- Students log in using their university credentials.
- Eligible voters are verified through the student database.

Process:

- Votes are cast through the user-friendly web interface.
- Each vote is recorded on the blockchain ledger, ensuring it is secure and tamper-proof.
- Votes are automatically counted in real-time.

Output:

- Accurate election results are generated immediately after voting ends.
- Administrators and students can verify the integrity of the results through the blockchain system.

XI. Expected Output

- Functional Web-Based Voting System (VoteChain)
- Blockchain-Based Vote Ledger
- Automated Election Reports
- Voter and Candidate Database

References

Aswini, T. (2025). Blockchain in electronic voting systems: trust and security challenges. Scientific Journal of Artificial Intelligence and Blockchain Technologies. <https://sjaibt.org/index.php/j/article/view/68>

Dhruthana, S., Chandana, P. Y., Inchara, B. S., Vandana, G. R., & Guruprasad, Y. K. (2025). Blockchain-based e-voting system. International Journal of Computer Techniques. <https://ijctjournal.org/blockchain-based-e-voting-system/>

Singh, S., Wable, S., & Kharose, P. (2022). A review of e-voting system based on blockchain technology. International Journal of New Practices in Management and Engineering. <https://www.ijnpme.org/index.php/IJNPME/article/view/125>